

Azure VNet Exercise

Part 1: Create Virtual Network (Foundation)

The image displays two screenshots from the Microsoft Azure portal. The top screenshot shows the 'Overview' page for a Virtual Network named 'vnet-demo-lab'. The page includes a left-hand navigation pane with options like Activity log, Access control (IAM), Tags, and Settings. The main content area shows 'Essentials' with details such as Resource group (rg1), Location (Central India), and Address space (10.0.0.0/16). Below this, the 'Capabilities' section shows various services like DDoS protection, Azure Firewall, and Microsoft Defender for Cloud, all marked as 'Not configured'.

The bottom screenshot shows the 'Add a subnet' configuration window. It prompts the user to select an address space and configure the subnet. The 'Subnet purpose' is set to 'Default', and the 'Name' is 'web-subnet'. Under the 'IPv4' section, the 'Include an IPv4 address space' checkbox is checked, and the 'IPv4 address range' is set to '10.0.0.0 - 10.0.255.255'. The 'Starting address' is '10.0.1.0' and the 'Size' is '/29 (8 addresses)'. The 'Subnet address range' is '10.0.1.0 - 10.0.1.7'. The 'IPv6' section is unchecked, indicating no IPv6 address ranges are present. The 'Private subnet' section provides information about private subnets and their security benefits. At the bottom, there are 'Add' and 'Cancel' buttons.

Microsoft Azure

Home > vnet-demo-lab-1775804489640 | Overview > vnet-demo-lab

vnet-demo-lab | Subnets

Virtual network

Search

+ Subnet Refresh Manage users Delete

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

Name	IPv4
web-subnet	10.0.1.0/29

Showing 1 subnet

Add or remove favorites by pressing Ctrl+Shift+F

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose: Default

Name: app-subnet

IPv4

Include an IPv4 address space: ☒

IPv4 address range: 10.0.0.0/16

Starting address: 10.0.2.0

Size: /29 (8 addresses)

Subnet address range: 10.0.2.0 - 10.0.2.7

IPv6

Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

Add Cancel

Give feedback

Microsoft Azure

Home > vnet-demo-lab-1775804489640 | Overview > vnet-demo-lab

vnet-demo-lab | Subnets

Virtual network

Search

+ Subnet Refresh Manage users Delete Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

Name	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
web-subnet	10.0.1.0/29	-	3	-	-	-
app-subnet	10.0.2.0/29	-	3	-	-	-

Showing 2 subnets

Add or remove favorites by pressing Ctrl+Shift+F

Successfully added subnet

Successfully added subnet 'app-subnet' to virtual network 'vnet-demo-lab'.

Give feedback

Part 2: Create Virtual Machines

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a low cost VM Help me choose the right VM size for my workload Help me create a VM optimized for high availability

Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription: Azure for Students

Resource group: rg1

Create new

Instance details

Virtual machine name: vm-web

< Previous Next: Disks > Review + create

Give feedback

 **Native SSH** ⓘ

MOST POPULAR

LOCAL MACHINE

Source machine

Source machine OS ⓘ

Windows

Source IP address ⓘ

Local IP | 115.244.89.221  [Connecting over a VPN?](#)

Destination VM

VM IP address ⓘ

Public IP | 135.235.195.149

VM port ⓘ

22

Connection prerequisites

VM access ⓘ

☐ Check inbound NSG rules

Check access

SSH command

Execute in your choice of local shell

ssh -i <private-key-file-path> azureuser@135.235.195.149 

 Private key file path missing [Edit settings](#)

Edit settings



```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd ../../
C:\>cd Users/gurno/Downloads

C:\Users\gurno\Downloads>ssh -i vm-web_key.pem azureuser@135.235.195.149
The authenticity of host '135.235.195.149 (135.235.195.149)' can't be established.
ED25519 key fingerprint is SHA256:EdryEkZFSBgflWpz2ofrJspWuDsmjnhsapum2DrWTVN0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '135.235.195.149' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Apr  9 06:10:00 UTC 2026

System load:  0.08               Processes:    130
Usage of /:   5.7% of 28.02GB    Users logged in: 0
Memory usage: 3%                IPv4 address for eth0: 10.0.1.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.
```

Create a virtual machine



Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose a VM size

Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * rg1

[Create new](#)

Instance details

Virtual machine name * vm-app

Region * (Asia Pacific) Central India

[Deploy to an Azure Extended Zone](#)

Availability options No infrastructure redundancy required

Security type Trusted launch virtual machines

[Configure security features](#)

[< Previous](#)

[Next : Disks >](#)

[Review + create](#)

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.

[Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * vnet-demo-lab

[Create new](#)

Subnet * app-subnet (10.0.2.0/29)

[Manage subnet configuration](#)

Public IP

[Create new](#)

Public IP addresses have a nominal charge. [Estimate price](#)

NIC network security group
☐ None
☒ Basic
☐ Advanced

Public inbound ports *
☐ None
☒ Allow selected ports

Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260409115014 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-202604... Start time : 9/4/2026, 12:05:50 pm

Subscription : Azure for Students Correlation ID : 0ed50fe5-5194-4da1-baee-56f3eb6911d9

Resource group : rg1

Deployment details

Next steps

Set up auto-shutdown Recommended

Monitor VM health, performance, and network dependencies Recommended

Run a script inside the virtual machine Recommended

Go to resource Create another VM Scale out your VM

Give feedback

Tell us about your experience with deployment

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure

Search resources, services, and docs (G+J)

Copilot

27BCS15716@ucd.in CHANDIGARH UNIVERSITY (CUC...

Home > CreateVm-canonical.ubuntu-24_04-lts-server-20260409115014 | Overview

vm-app Virtual machine

Help me copy this VM in any region Manage this VM with Azure CLI

Search

Help me copy this VM in any region

Connect Start Restart Stop Hibernate Capture Delete Refresh Scale Open in mobile Feedback CLI / PS

Essentials

Resource group (move) : rg1

Status : Running

Location : Central India

Subscription (move) : Azure for Students

Subscription ID : 5c9d415b-9260-4cf5-a6e3-b23b39d006d4

Operating system : Linux (ubuntu 24.04)

Size : Standard DC1s v3 (1 vcpu, 8 GiB memory)

Primary NIC public IP : 20.204.88.38

1 associated public IPs

Virtual network/subnet : vnet-demo-lab/app-subnet

DNS name : Not configured

Health state : -

Time created : 9/4/2026, 6:36 am UTC

Tags (edit) : Add tags

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine

Computer name : vm-app

Operating system : Linux (ubuntu 24.04)

VM generation : V2

VM architecture : x64

Networking

Public IP address : 20.204.88.38 (Network interface vm-app191)

1 associated public IPs

Public IP address (IPv6) : -

Private IP address : 10.0.2.4

Add or remove favorites by pressing Ctrl+Shift+F

Part 3: NSG (Network Security Group)

Home > Compute infrastructure | Virtual machines > vm-web

Compute infrastructure | Virtual machines

Search

Virtual machines Get started

Create Reservations

Virtual machines

Virtual Machine Scale Set (VMSS)

Compute Fleet

Arc Machines

Disks + images

Capacity + placement

Related services

Monitoring+Operations

Help

vm-web | Network settings

Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Connect

Bastion

Network settings

Load balancing

Application security groups

Network manager

Settings

Network security group vm-web-nsg (attached to networkInterface: vm-web618_x1)

Impacts 0 subnets, 1 network interfaces

Create port rule

Search rules

Source == all Destination == all Protocol == all

Action == all Port == all

Inbound port rules (5)

Prio...	Name	Port	Protocol	Source
300	SSH	22	TCP	Any
1000	vmport	80	TCP	Any
65000	AllowVnetInBound	Any	Any	VirtualNe
65001	AllowAzureLoadBalancerInB...	Any	Any	AzureLoa
65500	DenyAllInBound	Any	Any	Any

Outbound port rules (3)

Add or remove favorites by pressing Ctrl+Shift+F

```
azureuser@vm-web:~$ sudo ufw allow 'Nginx Full'
Rules updated
Rules updated (v6)
azureuser@vm-web:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
azureuser@vm-web:~$
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Part 4: Secure Network Traffic (Real Scenario)

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

```
azureuser@vm-app:~$ ping 115.244.89.221
PING 115.244.89.221 (115.244.89.221) 56(84) bytes of data.
^C
--- 115.244.89.221 ping statistics ---
32 packets transmitted, 0 received, 100% packet loss, time 31782ms

azureuser@vm-app:~$ ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=0.019 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=0.023 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=0.018 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=64 time=0.036 ms
64 bytes from 10.0.2.4: icmp_seq=5 ttl=64 time=0.034 ms
64 bytes from 10.0.2.4: icmp_seq=6 ttl=64 time=0.054 ms
64 bytes from 10.0.2.4: icmp_seq=7 ttl=64 time=0.055 ms
64 bytes from 10.0.2.4: icmp_seq=8 ttl=64 time=0.052 ms
64 bytes from 10.0.2.4: icmp_seq=9 ttl=64 time=0.051 ms
64 bytes from 10.0.2.4: icmp_seq=10 ttl=64 time=0.056 ms
^C
--- 10.0.2.4 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9245ms
rtt min/avg/max/mdev = 0.018/0.039/0.056/0.014 ms
azureuser@vm-app:~$ |
```